

Lecture FYS5419, April 3, 2024

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Discrete Fourier Transform (DFT)

$$|x\rangle = \vec{x} = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{n-1} \end{bmatrix}$$

each entry of output (after DFT)

$$y_k = \frac{1}{\sqrt{n}} \sum_{j=0}^{n-1} x_j e^{2\pi i (jk)/n}$$

$$|y\rangle = \frac{1}{\sqrt{n}} \begin{bmatrix} \sum_{j=0}^{n-1} x_j e^{2\pi i j \cdot 0/n} (=y_0) \\ \sum_{j=0}^{n-1} x_j e^{2\pi i j \cdot 1/n} \\ \vdots \\ \sum_{j=0}^{n-1} x_j e^{2\pi i j (n-1)/n} \end{bmatrix}$$

x_j are in general complex

$j = 0, 1, \dots, n-1$

y_k are also complex
 $k = 0, 1, \dots, n-1$

Example 6

$$|x\rangle = \{1, 2\} \quad x_0 = 1 \quad x_1 = 2$$

$$k=0 : y_0 = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \cdot 2 = 3/\sqrt{2}$$

$$k=1 : y_1 = \frac{1}{\sqrt{2}} + \frac{2}{\sqrt{2}} e^{i\pi} = -\frac{1}{\sqrt{2}}$$

$$|y\rangle = U|x\rangle = U \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$u^T u = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Small digression: Polynomial expansion

$$\begin{aligned} f(x_i) &\approx P_{n-1}(x_i) = P(x_i) \\ &= \sum_{j=0}^{n-1} a_j x_i^j = a_0 x_i^0 + a_1 x_i^1 + \dots \\ &\quad + a_{n-1} x_i^{n-1} \end{aligned}$$

$$|P\rangle = \begin{bmatrix} P(x_0) \\ P(x_1) \\ \vdots \\ P(x_{n-1}) \end{bmatrix} = V |a\rangle$$

$$V = \begin{bmatrix} 1 & x_0^1 & x_0^2 & \dots & x_0^{n-1} \\ 1 & x_1^1 & x_1^2 & \dots & x_1^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n-1}^1 & x_{n-1}^2 & \dots & x_{n-1}^{n-1} \end{bmatrix}$$

Van der Monde Matrix

This is similar to Fourier Transf

$$w = e^{2\pi i/n}$$

$$x_0^1 = e^{2\pi i \cdot 0 \cdot 1/n} = \underline{1}; \quad x_0^2 = e^{2 \cdot 2\pi i/n} = \underline{1}$$

$$x_1^1 = e^{2\pi i/n} = \omega$$

$$x_1^2 = e^{2 \cdot 2\pi i/n} = \omega^2$$

$$x_1^3 = \omega^3 \quad \dots \quad x_1^{n-1} = \omega^{n-1}$$

$$x_2^2 = e^{2 \cdot 2 \cdot 2\pi i/n} = \omega^4$$

$$U = \frac{1}{\sqrt{n}} \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & \omega & \omega^2 & \omega^3 & \dots & \omega^{n-1} \\ \vdots & \omega^2 & \omega^4 & \dots & \dots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & \omega^{n-1} & \omega^{(n-1)^2} & \dots & \dots & \omega^{(n-1)^2} \end{bmatrix}$$

$$U^{-1} = \frac{1}{\sqrt{n}} \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & \omega^{-1} & \dots & \dots & \omega^{-(n-1)} \\ \vdots & \omega^{-2} & \dots & \dots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & \omega^{-(n-1)} & \dots & \dots & \omega^{-(n-1)^2} \end{bmatrix}$$

orthogonality (unitarity)

$$(U U^{-1})_{jk} = \frac{1}{n} \sum_{l=0}^{n-1} \omega^j \omega^{-lk}$$

$$= \frac{1}{n} \sum_{l=0}^{n-1} \omega^{(j-k)l}$$

$$\omega = e^{2\pi i/n}$$

is equal to 1 only
if $j = k$.

Quantum Fourier transform (QFT)

n - qubits ; 2^n basis states

$$N = 2^n$$

$$|y\rangle = U|x\rangle \quad U U^\dagger = U^\dagger U = \mathbb{1}$$

This transformation preserves
orthogonality and norm

$$\langle x_i | x_j \rangle = \delta_{ij}$$

$$\begin{aligned}\langle y_i | y_j \rangle &= \langle x_i | \underbrace{U^\dagger U}_{\uparrow} | x_j \rangle \\ &= \langle x_i | x_j \rangle = \delta_{ij}\end{aligned}$$

The QFT is a linear operator that transforms an orthonormal basis into a new orthonormal basis.

Assume an original computational basis

$$\{ |0\rangle, |1\rangle, |2\rangle, \dots, |N-1\rangle \}$$

$N=4$; $n=2$ qubits

$$|00\rangle = |0\rangle \otimes |0\rangle =$$

$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$|01\rangle = |0\rangle \otimes |1\rangle =$$

$$\begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$|10\rangle = |1\rangle \otimes |0\rangle$$

$$|11\rangle = |1\rangle \otimes |1\rangle$$

our $N=4$ basis

$\{ |0\rangle, |1\rangle, |2\rangle, |3\rangle \}$

$$|j\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{i2\pi jk/N} |k\rangle$$

1-qubit case $N = 2^1 = 2$

$|0\rangle$ and $|1\rangle$

$$|\hat{0}\rangle = \frac{1}{\sqrt{2}} \sum_{k=0}^1 e^{2\pi i \cdot 0 \cdot k/2} |k\rangle$$

$$= \frac{1}{\sqrt{2}} [|0\rangle + |1\rangle]$$

$$|\hat{1}\rangle = \frac{1}{\sqrt{2}} \left[e^{2\pi i \cdot 0 \cdot 1/2} |0\rangle + e^{\pi i} |1\rangle \right]$$

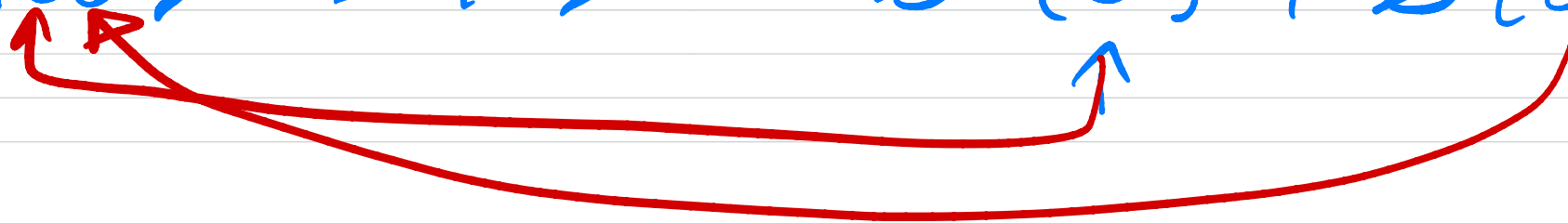
$$= \frac{1}{\sqrt{2}} [10\rangle - 11\rangle]$$

$$14\rangle = U \cdot 1x\rangle$$

$$U = H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$n = 2$ -qubit system, $N = 2^2 = 4$

possible encoding:

$$100\rangle \rightarrow 10\rangle : 2^1(0) + 2^0(0) = 0$$


$$|01\rangle \rightarrow |1\rangle : 2^{\overset{1}{1}0} + 2^0(1) = \underline{1}$$

$$|10\rangle \rightarrow |2\rangle : 2^{\overset{1}{1}0} + 2^0(0) = 2$$

$$|11\rangle \rightarrow |3\rangle : 2^{\overset{1}{1}0} + 2^0(1) = 3$$

$n = 3$ qubits $N = 8$ states

$$|y\rangle = |4\rangle = |100\rangle =$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} : 2^2(1) + 2^1(0) + 2^0(0) = 4$$

our transformed basis

$$|j\rangle = \frac{1}{\sqrt{N}} \sum_{g=0}^{N-1} e^{2\pi i j \cdot g/N} |g\rangle$$

$$g = \{ |0\rangle, |1\rangle, \dots, |N-1\rangle \}$$

$$[y_0 \ y_1 \ \dots \ y_{N-1}]$$

$$y = 2^{N-1} y_0 + 2^{N-2} y_1 + \dots + 2^1 y_{N-2} + 2^0 y_{N-1}$$

$$|j\rangle = \frac{1}{\sqrt{N}} \sum_{g_0=0}^1 \sum_{g_1=0}^1 \cdots \sum_{g_{n-1}=0}^1$$

$$\times \left\{ e^{2\pi i j / N} \sum_{k=0}^{n-1} g_k z^{n-k} \right\} |g_k\rangle$$

$$= \frac{1}{\sqrt{N}} \bigotimes_{k=0}^{n-1} \left[|0\rangle + e^{i 2\pi j z^{-k}} |1\rangle \right]$$

Note

$$|0\rangle + e^{i2\pi J} 2^{-k} |1\rangle \quad \left| \begin{array}{l} |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{array} \right.$$

$$= \begin{bmatrix} 1 \\ e^{i2\pi J/2^k} \end{bmatrix}$$

$$|J\rangle \rightarrow \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ e^{i2\pi J/2^0} \end{bmatrix} \otimes$$

$$\otimes \begin{bmatrix} 1 \\ e^{i2\pi j/2} \end{bmatrix} \otimes \dots \otimes \begin{bmatrix} 1 \\ e^{i2\pi j/2^{n-1}} \end{bmatrix}$$

Example:

$$n=2$$

$$|00\rangle \rightarrow \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} |00\rangle + |01\rangle \\ + |10\rangle + |11\rangle \end{bmatrix}$$

$$n=3$$

$$|001\rangle \rightarrow \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ e^{i2\pi/2} \end{bmatrix}$$

$$\otimes \begin{bmatrix} 1 \\ e^{i2\pi/4} \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ -1 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{8}} \left[\begin{aligned} &|000\rangle - |010\rangle + |100\rangle \\ &- |110\rangle + i(|001\rangle \\ &- |011\rangle + |101\rangle \\ &- |111\rangle) \end{aligned} \right]$$

For $n=2$, the QFT transformation

$$|x\rangle = \text{QFT}(n=2) |y\rangle$$

$$\frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{bmatrix}$$

$$|00\rangle \rightarrow \text{QFT} |00\rangle$$

$$\text{QFT}(n=2) \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

in general

$$\text{QFT}(N=2^n) =$$

$$\begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & \omega & \omega^2 & \dots & \omega^{N-1} \\ 1 & \omega^2 & \omega^4 & \dots & \omega^{2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{N-1} & \omega^{(N-1)^2} & \dots & \omega^{(N-1)^2} \end{bmatrix}$$

$$\omega = e^{2\pi i / N}$$

General state

$$|y\rangle = |y_0 y_1 \dots y_{n-1}\rangle$$

transforms in terms

$$\frac{1}{\sqrt{2^n}} (|0\rangle + e^{2\pi i y / 2^0} |1\rangle)$$

$$\otimes (|0\rangle + e^{2\pi i y / 2^1} |1\rangle) \otimes \dots$$

$$y = y_0 2^{n-1} + y_1 2^{n-2} + \dots + y_{n-1} 2^0$$

with binary float

$$O. y_k y_{k+1} \dots y_{n-1} =$$

$$y_k 2^{-1} + y_{k-1} 2^{-2} + \dots + y_{n-1} 2^{-(n-k)}$$

$$|y\rangle = |y_0 y_1 \dots y_{n-1}\rangle \rightarrow$$

$$\frac{1}{\sqrt{2^n}} (|0\rangle + e^{i2\pi O. y_{n-1}} |1\rangle) \otimes$$

$$(|0\rangle + e^{i2\pi O. y_{n-2} y_{n-1}} |1\rangle)$$

$$\otimes \dots \otimes (|0\rangle + e^{i2\pi O. y_0 y_1 \dots y_{n-1}} |1\rangle)$$

$$n=2$$

$$0. y_1 = y_1/2$$

$$0. y_0 y_1 = y_0/2 + y_1/4$$

$$|y\rangle = |y_0 y_1\rangle \rightarrow$$

$$\frac{1}{2} [\bar{10} + e^{2\pi i 0. y_1} 11] \otimes$$

$$[\bar{10} + e^{2\pi i 0. y_0 y_1} 11]$$

$$|00\rangle \rightarrow \frac{1}{2} (10 + 11) \otimes (10 + 11) \\ = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

DFT / QFT

$$|y\rangle = U|x\rangle$$

$$U \in (\mathbb{R}, \mathbb{C})^{n \times n}$$

$$O(n^2)$$

$$\text{FFT} \sim O(n \log n)$$